Multiple Object Tracking in Satellite Video With Graph-Based Multiclue Fusion Tracker

Haoxiang Chen, Nannan Li, Dongjin Li, Jianwei Lv, Wei Zhao, Rufei Zhang, and Jingyu Xu

***Abstract*— With the rapid advancement of satellite technology, satellite video has emerged as a key method for acquiring dynamic terrestrial information, facilitating multiple object tracking (MOT). Satellites are capable of surveying vast urban landscapes, yet the observed objects are small and dispersed among complex interference from the background, heightening the challenges in detection and association tasks for object tracking. However, current trackers often dissociate the classification task from the localization task, leading to drift in tiny object detection (TOD), and rely on prior knowledge for clue ranking, limiting model robustness. In this article, we introduce the graph-based multiclue fusion tracker (GMFTracker). Initially, we introduce a sparse sampling-based feature map correction approach to rectify the misalignment between the classification and localization feature maps. Furthermore, we developed graph neural networks (GNNs) for object relationship modeling, free from presuppositions, to tackle association challenges using relational features. GMFTracker was rigorously tested on VISO, CGSTL, and TinyPerson datasets, demonstrating its competitive performance relative to contemporary studies.**

***Index Terms*— Graph neural networks (GNNs), multiple object tracking (MOT), relational features, satellite video, task alignment, tiny object detection (TOD).**

I. Introduction

MULTIPLE object tracking (MOT) in satellite video plays a pivotal role in fields such as smart city initiatives, traffic management, and military reconnaissance. Significant advancements (see [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]) have been realized in this domain over recent years.

MOT in satellite videos comprises two distinct subtasks: tiny object detection (TOD) and object association (OA). These tasks necessitate the tracker's capability to precisely localize the position of tiny objects in each frame and to associate the detection results across successive frames. However, MOT in satellite videos remains unsolved due to the following challenges.

1) *Extreme Imbalance Between Foreground and Background:* Objects shown in satellite bird's-eye views usually occupy less than 40 pixels, which severely increases the false negatives of the detector.
2) *Significant Interference:* Much of the interference reflects as a white spot of light similar to the target, which enhances the false positives of the detector.
3) *Crowded Targets:* Tiny spacing between targets presents an association risk, leading to broken trajectories for the objects.

Therefore, the current MOT in satellite video has attracted numerous scholars to study in recent years.

Widely utilized MOT algorithms (see [11], [12], [13], [27]) enhance the multiscale feature representation of targets, thereby improving the detection of tiny objects. However, this approach not only increases the computational burden but also overlooks the issue of localization drift in TOD. In conjunction with the conclusion of [13], it is discerned that the essential cause of localization drift is task misalignment, which causes the detection box to display the maximum intersection over union (IoU) with the ground truth (GT) does not consistently exhibit the highest confidence score. Specifically, with extremely small target sizes, minor positional deviations can significantly impact the IoU, which can significantly reduce the detection accuracy.

During the association phase of MOT, numerous pairs of relational indicators (clues) are utilized to determine whether the detection results from adjacent frames belong to the same target. Most existing trackers (see [14], [15], [16], [17], [18], [19]) utilize the appearance similarity, IoU, and center distance between two frames for association. Zhang et al. [14], Wang et al. [15], Zhang et al. [20], Cao et al. [21], and Zhang et al. [22] hypothesized the significance of appearance, motion, and positional clues based on a priori knowledge. However, these methods present clear limitations: the necessity to depend on empirical or experimental results to ascertain the importance of each clue, which undermines the model's robustness in unfamiliar scenarios.

In this article, a graph-based multiclue fusion tracker (GMFTracker), including the tiny object task correction (TOTC) module and the local scene multiclue integration (LSMcI) module, is proposed to solve the above difficulties. TOTC employs feature correction to compensate the offset between the classification task and the localization task to improve the detection accuracy of tiny objects. Meanwhile, the idea of sparse sampling is used to make local self-attention to improve the inference speed. Furthermore, to address the issue

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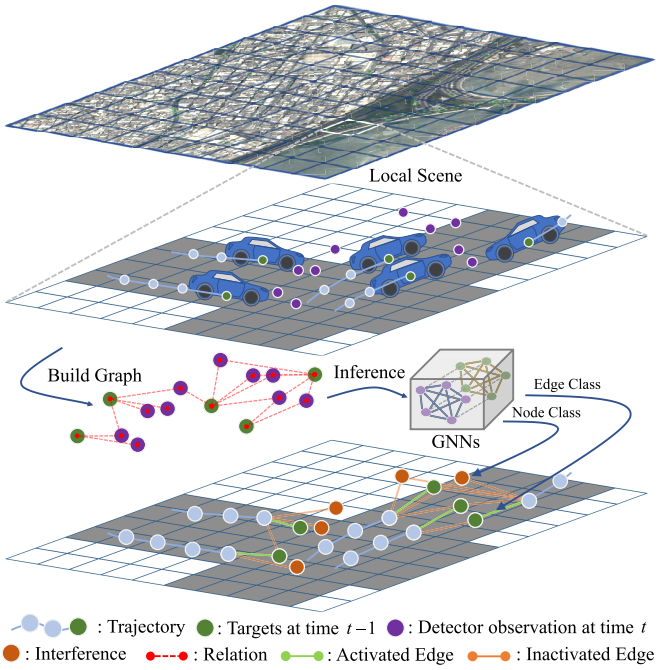


Fig. 1. OA structure based on GNNs. Initially, the relationship features are constructed by utilizing green nodes and purple nodes. Subsequently, these features are propagated and aggregated via GNNs. Ultimately, the association task is executed using the activation or inactivation of edge classifiers, and the interference rejection task is executed by the node classifier.

of dependence on prior assumptions, we propose the LSMcI. As shown in Fig. 1, the LSMcI first utilizes graph neural networks (GNNs) to pass and aggregate information between neighboring objects. Subsequently, LSMcI transforms the OA task into an edge classification task and the interference rejection task into a node classification task, respectively. Specifically, this approach allows the model to decide the significance of each clue based on the neighboring targets, rather than a predefined rule. For example, if the appearance similarity and IoU of neighboring targets are very close, then the motion direction and aspect ratio will be the main basis of association.

In summary, the main contributions are as follows.

1) We propose an end-to-end tracking network, designated as GMFTracker, for precise localization and enhanced association robustness, to achieve real-time MOT of satellite videos.
2) In the TOTC module, the strategy of feature correction is proposed to alleviate the misalignment between the classification and localization tasks. Moreover, the strategy of sparse sampling is proposed to reduce the computational overhead. In this way, the detection accuracy of tiny objects is improved effectively.
3) The LSMcI module is proposed to dynamically infer the optimal combination of association clues via GNNs without priori. Subsequently, LSMcI recovers the detection based on the association results of the GNNs and rejects the interference by node feature, which improves the accuracy of the association.

Experiments are performed on three benchmark datasets, namely VISO [1], CGSTL [2], and TinyPerson [52]. Through comprehensive experiments and comparisons with recent state-of-the-art (SOTA) methodologies, we confirm the efficacy of the proposed methodology.

II. Related Work

This section briefly reviews the recent popularity of TOD and satellite video-based MOT. In addition, since GMFTracker utilizes GNNs to infer relational features, we will review the related basic model.

A. Tiny Object Detection

Tiny objects occupy a small portion of the image and have almost no texture features, which makes the feature representation of the model very difficult. Existing methods enhance the feature representation mainly in terms of scale-aware and attention-aware.

At the level of scale-aware, multiscale deep CNN [25] employs deconvolution operations on feature maps of various scales to enhance their respective resolutions. Then, these scale-enlarged feature maps are used for prediction. TridentNet [26] utilizes a parallel multibranch architecture, aiming to manage the optimal receptive field for objects of varying scales, thus facilitating the network's adaptation to smaller-scale objects. QueryDet [11] utilizes a multiscale feature map querying approach to enhance small object detection, thereby diminishing superfluous computations in low-level features. SSPNet [12] implements a feature-sharing strategy in FPN, capitalizing on the relations between feature maps of adjacent levels, thereby addressing the challenge of inconsistent gradients across feature maps at varying scales. At the level of attention assistance, FBRNet [27] integrates an attention mechanism into FCOS [28], balancing feature maps across various levels of the feature pyramid and bolstering the learning efficacy for small objects. To obtain stronger responses for small objects, CANet [29] designs a cross-layer attention module. While the above approach wisely allocates additional computational resources to enhance the features of tiny objects, it ignores the problem of localization drift due to task decoupling.

B. Satellite Video Object Tracking

Visual tracking in remote sensing has developed rapidly in recent years, which can be roughly summarized into joint detection tracking (JDT) and detection-based tracking (DBT).

In terms of JDT, CFTracker [3] proposes a cross-frame feature update (CFU) module and a cross-frame training flow (TC) policy, respectively. CFU and TC can extract semantic information across frames, which enable the network to yield motion prediction results characterized by enhanced temporal consistency. This advancement contributes to diminishing ID flickering and minimizing false detections. OKD-JDT [7], utilizing adaptive attention learning, facilitates efficient knowledge distillation, thereby ensuring precise tracking within low-resolution imagery. STARK [8] proposes the bi-direction propagation and fusion (Bi-PF) strategy for distinctive representation and target-aware enhancement (TAE) module for augmented target search, resulting in superior tracking performance in satellite videos. In terms of DBT, BMTC [5]

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approaches MOT in satellite videos as a parallelization of multiple single-object tracking tasks. Furthermore, it introduces trajectory criteria and employs the invalid fragment trajectory backtracking (IFTB) technique to tackle challenges related to trajectory fragmentation and continuity. To augment tracking performance, MBLT [4] utilizes a rapid clustering approach to segment feasible regions, subsequently amalgamating network predictions regarding the probability of each pixel's presence in the subsequent frame, indicative of the likelihood of object occurrence. DFSNet [6] presents a dual-stream network featuring dynamic and static fusion, which extracts static clues from single frames and dynamic clues from consecutive frames, subsequently integrating a tracker to implement a DBT scheme. CSLT [9] utilizes spatial-attention fusion for effective feature integration and a model drift alarm and calibration (MDC) module, ensuring robust tracking of dim objects in satellite videos. Furthermore, it mitigated static false alarms by integrating smoothed track discontinuities into the Kalman filter algorithm. While the aforementioned methodologies have achieved advancements in enhancing tracking performance, the majority employ dual-stream networks, intricate association logics, and combinations of association clues based on a priori knowledge. Such selections result in diminished processing speeds and constrain the robustness of association algorithms.

C. GNN Modeling Relations

GNNs were systematically described by Zhou et al. [30], processing data characterized by graph structures using neural networks. In this paradigm, objects and relations are precisely modeled as nodes and edges within the graph structure, interconnected in a distinct manner.

GNNs have a unified paradigm for modeling relationships, and the graph is defined as a binary $G(V, E)$, where $V = \{n_i\}$ is the set of nodes, each n_i being an attribute feature of the ith node, $E = \{(e_k, r_k, s_k)\}$ is the set of edges, each e_k being an attribute feature of the kth edge, with r_k and s_k the indexes of the receiving and sending nodes of the edges, respectively. The computational unit of GNNs is a module that inputs a graph and returns a graph

$$e'_k = \phi^e\left(e_k, n_{r_k}, n_{s_k}\right) \tag{1}$$

$$\bar{e}'_i = \rho^{n\to e}\left(E'_i\right) \tag{2}$$

$$n'_i = \phi^n\left(\bar{e}'_i, n_i\right) \tag{3}$$

where $E'_i = \{(e'_k, r_k, s_k)\}_{r_k=i,k=1:N^e}$ and ϕ and ρ are update and aggregation functions that can be modeled by neural networks, respectively. The output graph of this computational unit is defined as follows:

$$G' = \left(V', E'\right) \tag{4}$$

$$V' = \left\{n'_i\right\}_{i=1:N^v} \tag{5}$$

$$E' = \cup_i E'_i \tag{6}$$

where N^v and N^e are the total number of nodes and edges in the graph, respectively. In summary, $G(V, E)$ completely models the relationship between individuals and collectives in a scene and retains a strong physical meaning. This inherent property has led to an increasing amount of in-depth scholarly research on GNNs (see [10], [15], [24], [31], [38]).

SUSHI [31] employs a hierarchical graph structure to manage data associations across various temporal extents in MOT, leveraging the node relationships within the graph to preserve both long-term and short-term target identities. Furthermore, MPNTrack [24] conceptualizes the association model as a maximum flow issue, updating the message passing network [38] with time-aware nodes to effectively integrate past and future features. STeRe [10] reinterprets the tuple set adjacency matrix, representing each frame of the video as a fully connected graph to encapsulate the similarity information effectively. GSDT [15], alternatively, represents feature maps within regions of interest of consecutive frames as nodes, aimed at managing computational resources efficiently. This approach also enables GNNs to learn more distinctive spatiotemporal object relationships.

III. Method

In this section, the proposed GMFTracker is introduced, and first, we introduce the overall structure of the network in Section III-A. Then, the TOTC for task alignment is presented in Section III-B. Next, we introduce the target association method LSMcI without a priori assumptions in Section III-C. Finally, the loss function is presented in Section III-D.

A. Overview

MOT aims to interpret observations using a collection of trajectories $\Lambda_* = \{L_0, \ldots, L_k, \ldots, L_m\}$, where m denotes the count of traces in a video. Given a satellite video sequence $I = \{I_1, \ldots, I_t, \ldots, I_n\}$, where I_t denotes the tth frame in the video, the video is extracted with a set of detection results $D = \{D_1, \ldots, D_t, \ldots, D_n\}$, where D_t denotes the detection results at the tth frame and $D_t = \{o_t^1, \ldots, o_t^i, \ldots, o_t^{n_t}\}$ with o_t^i denoting the detection result of the ith object at the tth frame of the video. There are n_t targets detected at tth frame. Each detection result is described by a tuple with ten elements $o_t^{n_t} = [t, \mathrm{id}, x_c, y_c, w, h, s, c, a, f]$, where t is the frame number; id is the unique identity of the object; x_c and y_c denote the coordinates of the center of the detection box; w and h denote the width and height of the detection box, respectively; s denotes the confidence score; c denotes the category of the object; a denotes the direction of motion; and f denotes the feature embedding of the target.

As shown in Fig. 2, GMFTracker has two inputs, $\{\cdot\}_t$ and $\{\cdot\}_{t-1}$, where $\{\cdot\}_t$ is composed of three images from I_t, I_{t-1}, and I_{t-2}; if $t - N$ be less than 0, then I_{t-N} equals I_0. In the initial data preprocessing phase, the images on $\{\cdot\}_t$ are concatenated along the channel dimension, resulting in an input tensor with the size of $H_{\mathrm{img}} \times W_{\mathrm{img}} \times 9$, where H_{img} and W_{img} are the height and width of the image, respectively. The above operation also applies to $\{\cdot\}_{t-1}$. We employ the deep layer aggregation (DLA-34) backbone network, as described in the literature [19]. The features output from the backbone network are fed into a simple temporal feature fusion (TFF) module, which is described as follows:

$$F_{t\oplus t-1} = \mathcal{F}(F_t, F_{t-1}) \tag{7}$$

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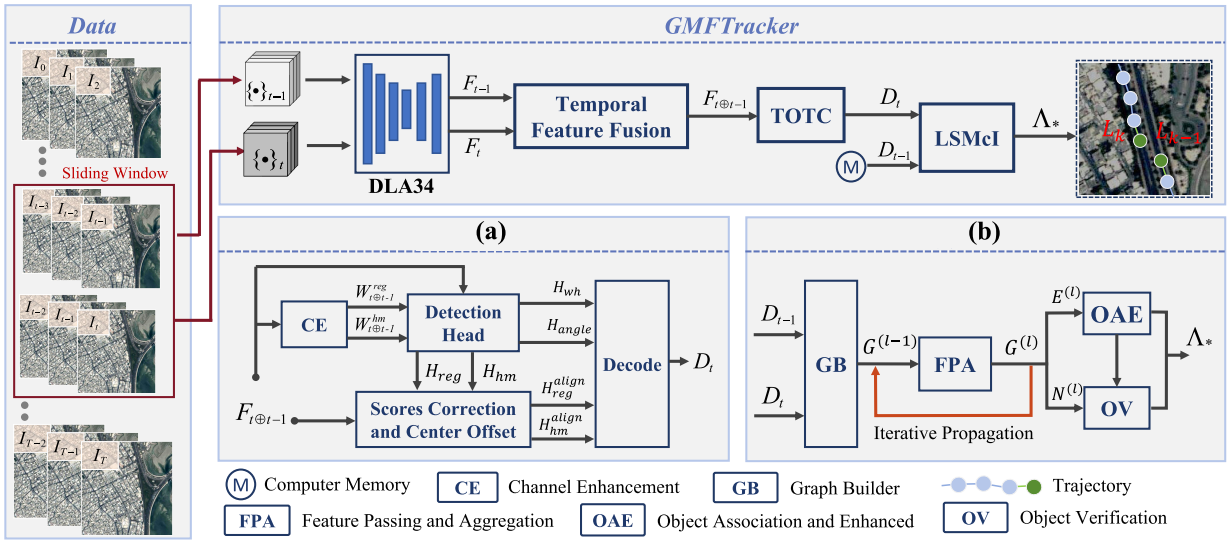


Fig. 2. Overview of our GMFTracker architecture. Initially, two sets of image collections, $\{\cdot\}_t$ and $\{\cdot\}_{t-1}$, are obtained from the image sequence, and their features are extracted using the DLA34. After TFF, $F_{t\oplus t-1}$ obtains detection results and node features via TOTC. Subsequently, the detection results and features are input into LSMcI to accomplish the association task by aggregating higher order clues from adjacent targets. Ultimately, OAE and OV implement association enhancement and interference rejection operations to maintain the trajectory set Λ_*.

where $\mathcal{F}$ denotes a classical feature fusion compression method, variational encoder (VAE), which is composed of fully connected layers, and F_t and F_{t-1} denote the backbone features extracted from two image sets, $\{\cdot\}_t$ and $\{\cdot\}_{t-1}$, respectively. The features $F_{t\oplus t-1} \in \mathbb{R}^{B\times H\times W\times C}$, emanating from the TFF, encompass spatiotemporal semantic information, where $C = 64$, $H = H_{\text{img}}/4$, and $W = W_{\text{img}}/4$.

In the detection phase, $F_{t\oplus t-1}$ enters the TOTC to implement feature correction. First, channel enhancement (CE) selectively enlarges the characterization of classification and localization features in different channels utilizing $W^{hm}_{t\oplus t-1}$ and $W^{\text{reg}}_{t\oplus t-1}$. Afterward, Scores Correction and Center Offset (SCCO) corrects the classification and localization features based on a sparse sampling strategy to obtain H^{align}_{hm} and $H^{\text{align}}_{\text{reg}}$. Finally, the corrected features are passed through detection head and decoder to get the detection result D_t.

In the tracking phase, D_t and D_{t-1} enter the LSMcI to achieve the OA. First, graph builder (GB) will construct edge features $e^{(l)} \in E^{(l)}$ and node features $n^{(l)} \in V^{(l)}$ based on the detection results, where l denotes the number of iterations. After that, feature passing and aggregation (FPA) diffuses the information of neighboring targets to each other. As the number of FPA iterations increases, the information diffuses more widely, and eventually, $e^{(l)}$ and $n^{(l)}$ become one-to-many association clues.

Inspired by the network maximum flow model [45] and the method of maximum log likelihood, LSMcI models the OA problem as follows:

$$\max_{y} \sum_{i} \sum_{j} c_{i,j}(D_t, D_{t-1}, G_t, G_{t-1}) \cdot y_{i,j} \tag{8}$$

$$\max_{\theta} \sum_{i} \sum_{j} \log P_\theta\left(\tilde{y}_{i,j} | e^{(l)}, n^{(l)}\right). \tag{9}$$

Equation (8) is the basic formula for network maximum flow, where $c_{i,j}$ is the benefit value, which is determined from the detection results (D_t and D_{t-1}) and GT (G_t and G_{t-1}). $y_{i,j} \in \{0, 1\}$ denotes whether the edge is activated or not, and according to (8), the association problem is transformed into an edge activation problem. Equation (9) uses the concept of maximum log likelihood to transform the edge activation problem into an edge classification problem by softening $y_{i,j}$ to $\tilde{y}_{i,j} \in [0, 1]$, which denotes the probability of classification. The OA in object association and enhanced (OAE) is an edge classification model with parameter θ, which estimates the likelihood of an edge being activated by $e^{(l)}$. Moreover, the object enhancement (OE) in OAE is able to recover part of the detection boxes based on the association results, while object verification (OV) as a node classifier is able to reject the recovered interferences. Algorithm 1 details the specific processes of OA, OE, and OV.

B. TOTC Module

The tasks of object classification and localization are associated with distinct truth-value encodings and loss functions, necessitating that these two branches concentrate on divergent feature types. Nevertheless, the single-branch architecture of the backbone compels disparate tasks to utilize a shared feature map. As described in the literature [41], distinct channels of feature mapping can learn different descriptors for distinct tasks. Thus, our aim is to develop a module that selectively amplifies specific feature channels for distinct tasks. In addition, two distinct sets of learnable correction quantities are formulated to address the offset between the output features of classification and localization models. Inspired by TOOD [13] and CBAM [42], we propose the TOTC to enhance the task alignment capability of detectors in tiny object scenarios. As shown in Fig. 3, TOTC consists of two parts: CE and SCCO. The former facilitates task decomposition by dynamically calculating task-specific features within the channels, whereas the latter enhances classification and localization tasks

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by modulating the spatial distribution of the two correction quantities, $S^c_{t\oplus t-1}$ and $S^p_{t\oplus t-1}$. The ablation experiments in Section IV-D demonstrate the effectiveness of TOTC.

The structural details of the TOTC are illustrated in Fig. 3. Initially, a specified feature $F_{t\oplus t-1}$ is augmented by the CE to output $W^{\text{reg}}_{t\oplus t-1}$ and $W^{hm}_{t\oplus t-1}$. Subsequently, $F_{t\oplus t-1}$ is processed through the conventional feature detection heads, resulting in H_{hm}, H_{reg}, H_{wh}, and H_{angle}. Here, H_{hm} and H_{reg} are refined using the outputs of the SCCO, $S^c_{t\oplus t-1}$ and $S^p_{t\oplus t-1}$, to yield H^{align}_{hm} and $H^{\text{align}}_{\text{reg}}$.

In CE, $F^{\max}_{t\oplus t-1}$ and $F^{\text{avg}}_{t\oplus t-1}$ represent the results derived from maximum pooling and average pooling operations. Subsequently, $F^{\max}_{t\oplus t-1}$ and $F^{\text{avg}}_{t\oplus t-1}$ are processed through a semi-shared MLP, comprising three fully connected layers, with only one layer being shared. The outputs of semi-shared MLP are subsequently aggregated and collectively input into a sigmoid function to derive two distinct sets of weights, $W^{\text{reg}}_{t\oplus t-1}$ and $W^{hm}_{t\oplus t-1}$. The calculation process is as follows:

$$f^*_{\text{ce}}(x, y) = \text{sMLP}^*(\text{MaxPool}(x) + \text{AvgPool}(y)) \tag{10}$$

$$W^{hm}_{t\oplus t-1} = \sigma\left(f^1_{\text{ce}}(F_{t\oplus t-1}, F_{t\oplus t-1})\right) \tag{11}$$

$$W^{\text{reg}}_{t\oplus t-1} = \sigma\left(f^2_{\text{ce}}(F_{t\oplus t-1}, F_{t\oplus t-1})\right) \tag{12}$$

where sMLP* denotes semi-shared MLP, which indicates that only the first fully connected layer shares weights. σ denotes the sigmoid function.

A constant term of 1 is appended to each set of weights prior to the initiation of feature augmentation. This step ensures that the initialization is identical, whether or not CE enhancement has been performed.

In SCCO, $F_{t\oplus t-1}$ executes a sequence of operations: convolution, batch normalization, channel compression, and nonlinear processing, in the specified order. In addition, it is noteworthy that the two branches operate independently, with no shared weights. The module yields confidence corrective feature $S^c_{t\oplus t-1} \in \mathbb{R}^{B\times H\times W\times N}$ and spatial offset feature $S^p_{t\oplus t-1} \in \mathbb{R}^{B\times H\times W\times 8}$ as outputs. Channel compression is imperative in this context, as $S^c_{t\oplus t-1}$ and $S^p_{t\oplus t-1}$ represent the feature weights for categorization and the offsets for localization, respectively. In short, the SCCO is computed as

$$S^c_{t\oplus t-1} = \mathbf{W_2}(\mathbf{W_1}(F_{t\oplus t-1})) \tag{13}$$

$$S^p_{t\oplus t-1} = \mathbf{W_4}(\mathbf{W_3}(F_{t\oplus t-1})) \tag{14}$$

where $\mathbf{W_1}$ and $\mathbf{W_3}$ represent convolution operations with kernel size of 3×3, respectively, and $\mathbf{W_2}$ and $\mathbf{W_4}$ represent convolution operations with kernel size of 1×1.

Specifically, $S^c_{t\oplus t-1}$ facilitates the adjustment of the confidence score at each spatial location, while $S^p_{t\oplus t-1}$ facilitates the identification of the optimal sampling point in its vicinity by the center point. As illustrated in Fig. 3, we have developed the dynamic offset layer (DOL), drawing inspiration from the DCN and self-attention. The computational procedure for H^{aling}_{hm} and $H^{\text{align}}_{\text{reg}}$ following task alignment is detailed as follows:

$$H^{\text{aling}}_{hm} = \sqrt{H_{hm} \times S^c_{t\otimes t-1}} \tag{15}$$

$$\begin{aligned} H^{\text{align}}_{i,j} &= Dy - \text{offset}(x_p, p_{i,j}, H_{\text{reg}}) \\ &= \sum_{\Delta p_n \in S_{i,j}} x_p \cdot \left[\mathbf{W}^1_d H_{\text{reg}}\left(p_{i,j} + \Delta p_n\right)\right]^T \\ &\quad \cdot \mathbf{W}^2_d H_{\text{reg}}\left(p_{i,j} + \Delta p_n\right) \end{aligned} \tag{16}$$

where $[\cdot]$ denotes concatenation; $x_p = H_{\text{reg}}(i, j, :) \in \mathbb{R}^{1\times 1\times C}$; $p_{i,j}$ denotes the reference point of the feature map; $S_{i,j} = S^p_{t\oplus t-1}(i, j, :) \in \mathbb{R}^{1\times 1\times 8}$ represents the location of the four offset points; Δp_n signifies the deviation from $P_{i,j}$; and $\mathbf{W}^1_d$ and $\mathbf{W}^2_d$ denote the two sets of learnable weights, respectively. Equation (16) demonstrates that DOL significantly diminishes the computational load while executing the optimal offset operation via sparse sampling. Through Section IV-F, the sparse sampling method is essential to achieve effective online tracking of the tracker.

C. LSMcI Module

Three prevalent characteristics are observed in many of the current MOT networks addressing association tasks: 1) dependence on object appearance, texture, and shape features in adjacent frames for association (see [15], [43], [44]); 2) evaluating and weighting each association clue according to its significance based on empirical evidence and experimental trials (see [14], [15], [21], [44]); and 3) focusing on one-to-one association clues, disregarding one-to-many complexities (see [21], [43], [44]). However, in satellite video, the aforementioned characteristics manifest as limitations. Tiny objects are devoid of distinct textures and exhibit similarity, thereby making distinction difficult; intricate road environments exacerbate the uncertainty of clues prioritization; and one-to-one association clues are not conducive to precise matching in crowded scenes.

We introduce a new association module, LSMcI, as depicted in Fig. 4. This module treats MOT association issues as a graph edge classification problem, utilizing GNNs for higher order relational features across multilevel jumps. The LSMcI offers two key benefits: 1) there is no need to arrange association clues in accordance with prior knowledge and 2) aggregation of relational features from nearby detections in adjacent frames, resulting in one-to-many association clues through GNNs' inference. MOT aims to interpret observations using a collection of trajectories $\Lambda_* = \{L_0, \ldots, L_k, \ldots, L_m\}$, where m denotes the count of traces in a video. Within a target tracking scenario, every node $i \in V$ signifies a detection result $o^i_t \in D_t$, produced by the detector at time t. Then, the target's trajectory can be described as $L_i = \{o^i_1, \ldots, o^i_t, \ldots, o^i_T\}$, which is mapped to nodes of graph G as $\{i^1, \ldots, i^t, \ldots, i^T\}$. Thus, the set of represented edges is $\{(i^1, i^2), \ldots, (i^{T-1}, i^T)\} \subset E$, and we define a binary variable $y(i, j) \in Y$ as

$$y(i, j) = \begin{cases} 1, & \exists L_k \in \Lambda_* \text{ s.t. } \left(o^i_{t-1}, o^j_t\right) \in L_k \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

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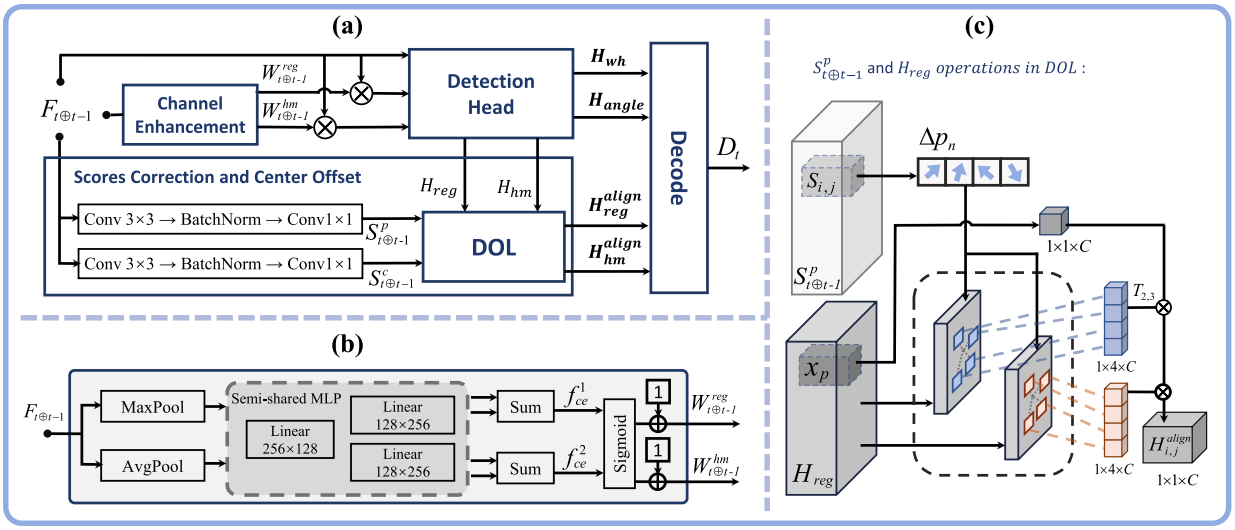


Fig. 3. Schematic of (a) TOTC module. The temporally fused features go through (b) CE to facilitate task interaction, subsequently integrating with category score correction and centroid offset to accomplish task alignment. In particular, (c) DOL integrates the sparse sampling idea to realize the alignment of the two tasks.

Inspired by the flow conservation constraint in [45], under the assumption that characterizing two targets with identical observations is infeasible, and nodes comprising each trajectory within the trajectory L_* are nonoverlapping. In executing an edge classification utilizing GNNs, each $y(i, j)$ is assigned a specific benefit $c(i, j)$. This benefit is pivotal in the edge classification task, dictating edge activation and influencing the optimal partition derived through optimization. Final partitioning is achieved via optimization

$$\max_{y} \sum_{(i,j)\in E} c(i,j)y(i,j) \tag{18}$$

$$\text{s.t.} \sum_{(i,j)\in E} y(i,j) \le 1$$

$$\left(o_{t-1}^{i}, o_{t}^{j}\right) \in L_k$$

$$y(i,j) \in \{0,1\}, \quad (i,j) \in E. \tag{19}$$

Drawing on the concept of maximum likelihood, $y(i, j)$ is softened by $\tilde{y}(i, j) \in [0, 1]$, and referring to (9), the objective function can be reformulated as

$$\max_{\theta} \sum_{(i,j)\in E} \log P_\theta\left(\tilde{y}(i,j)|e_{i,j}^{(l)}, n_i^{(l)}, n_j^{(l)}\right). \tag{20}$$

As shown in Fig. 4, the inputs of the LSMcI are D_t from TOTC and D_{t-1} from memory. Initially, the edges and nodes of the graph G are initialized using D_t and D_{t-1}. After that, G undergoes processing through FPA, facilitating the execution of feature transfer and feature aggregation operations. Ultimately, the node features $n_i^{(l)} \in V^{(l)}$ and edge features $e_{i,j}^{(l)} \in E^{(l)}$ are directed to the OAE and OV, where l denotes iterations of iterations and $l \ge 0$, and i and j denote the start and end nodes, respectively, with i originating from b_t and j from b_{t-1}.

In GB, initializing node features $n_i^{(0)}$, edge features $e_i^{(0)}$, and the graph itself is required. Node features $n_i^{(0)}$ are derived from the feature vectors corresponding to the central points of each detection frame in $F_{t\oplus t-1}$. For the edge features, following the approach in [24], we initialize the high-dimensional edge features as follows:

$$e_{i,j}^{(0)} = f_e\left(\left[x_i - x_j, y_i - y_j, \log\left(\frac{w_i}{w_j}\right), \log\left(\frac{h_i}{h_j}\right), \text{IoU}_{i,j}, \text{Sim}_{i,j}, \text{Ddm}_{i,j}\right]\right) \tag{21}$$

$$\text{Ddm}_{i,j} = a_i - \text{atan2}(x_i - x_j, y_i - y_j) \tag{22}$$

where x and y represent the center coordinates of the detection box; h and w are the height and width of a detection box, respectively; Sim denotes cosine similarity; Ddm denotes the distance in the direction of motion; a denotes the predicted direction of motion; and $f_e(\cdot)$ refers to two-layer fully connected network. It should be noted that the initialization features of the edges are direction-aware, due to the different encoding of the two reverse direction features for $i \to j$ and $j \to i$.

To initialize the graph $G^{(0)} = (V^{(0)}, E^{(0)})$, we need the set of node features $V_{t-1}^{(0)} = \{n_i^{(0)} | 1 \le i \le K\}$ at the moment $t-1$, the set of node features $V_t^{(0)} = \{n_j^{(0)} | 1 \le j \le K\}$ at the moment t, and the nodes that failed to be associated at previous moments $V_{\text{miss}}^{(0)} = \{n_k^{(0)} | 1 \le k \le |V_{\text{miss}}^{(0)}|\}$, combining all sets as follows:

$$V^{(0)} = V_{t-1}^{(0)} \cup V_t^{(0)} \cup V_{\text{miss}}^{(0)} = \left\{n_m^{(0)} | 1 \le m \le 2K + \left|V_{\text{miss}}^{(0)}\right|\right\} \tag{23}$$

$$E^{(0)} = \left\{e_{i,j}^{(0)} | 1 \le i, j \le 2K + \left|V_{\text{miss}}^{(0)}\right|\right\}. \tag{24}$$

In satellite video contexts, the excessive number of instances causes computational overload. Consequently, node connections mainly occur in localized regions. For node connectivity at the moment $t-1$ to t, two connection criteria are pivotal: 1) minimal center distance and 2) higher IoU score. For each criterion, the top K nodes are selected for connection, and duplicate connections are avoided.

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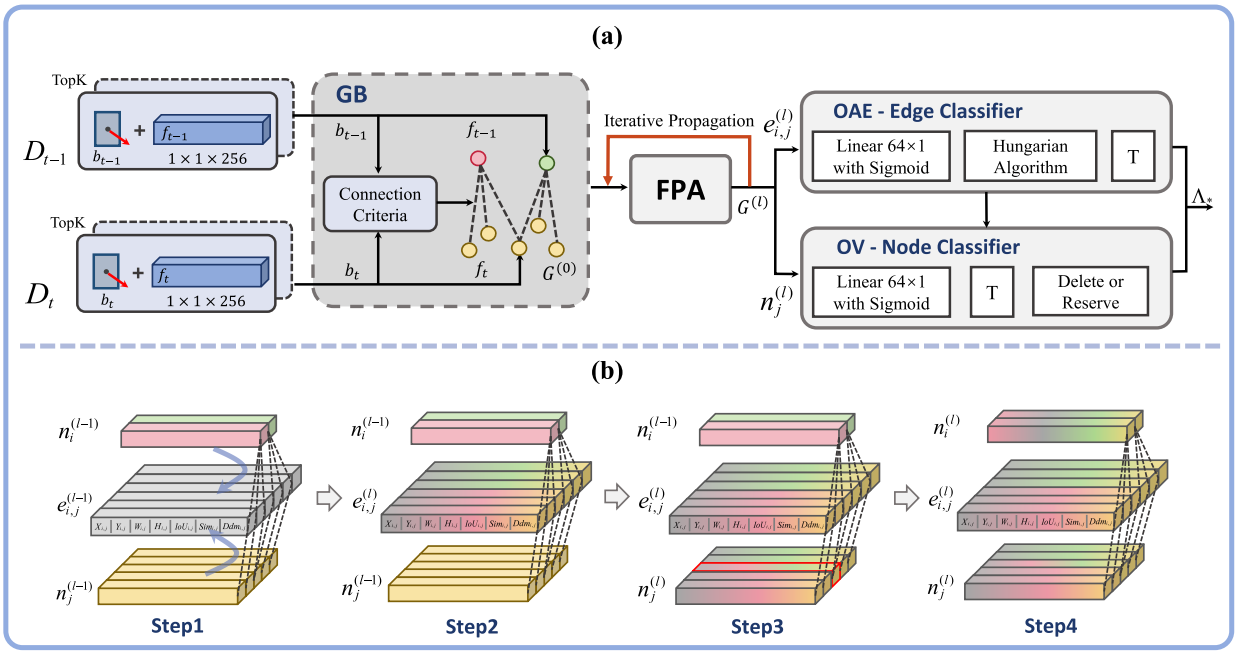


Fig. 4. Schematic of LSMcI module. LSMcI constructs the graph from detection results and feature vectors, aggregates multiclue features using GNNs, and completes the association task and interference rejection with edge and node classifiers. (a) LSMcI. (b) FPA.

Algorithm 1 Local Scene Multiclue Integration

Input: the output of TOTC at the moment t, which is decoded as $o_t^i = [t, id, x_c, y_c, w, h, s, c, a, f]$, $o_t^i \in D_t$ and D_{t-1} in memory.

Output: updated D_t and Y.

for all $t = 1, 2, \ldots, T$ **do**
  Initialize D_t and D_{t-1} to get graph G.
  for all $l = 1, 2, \ldots, N_{iter}$ **do**
    Message from V to E via Equation (25).
    Message from E to V via Equation (26).
  end for
  V and E are scored by the classifier, outputs $S_V \in \mathbb{R}^{n_t}$ and $S_E \in \mathbb{R}^{n_t \times n_{t-1}}$.
  Hungarian algorithm effect S_E gets the result Y.
  for all each edge from E **do**
    Retain o_t^i **if** $s \geq \tau_V$ and $S_E(i, j) \geq \tau_E$,
    Retain o_t^i **elif** $s < \tau_V$ and $S_E(i, j) \geq \tau_E$ and $S_V(j) \geq \tau_V$,
    else Eliminate interference o_t^i and $y(i, j) := 0$.
  end for
  Update D_t,store D_{t-1} and Y.
end for

FPA achieves feature passing and aggregation through embedding updates to edges and nodes as follows:

$$e_{i,j}^{(l)} = \text{avg}\left(f_{\text{edge}}\left(\left[n_i^{(l-1)}, n_j^{(l-1)}, e_{i,j}^{(l-1)}\right]\right), e_{i,j}^{(0)}\right) \tag{25}$$

$$n_j^{(l)} = \text{avg}\left(f_{\text{node}}\left(\frac{1}{\left|E_{:,j}^{(l)}\right|}\sum_i f_{\text{agg}}\left(\left[n_i^{(l-1)}, e_{i,j}^{(l)}\right]\right)\right), n_j^{(0)}\right) \tag{26}$$

where $f_{\text{edge}}(\cdot)$ and $f_{\text{node}}(\cdot)$ are the two-layer fully connected network, $f_{\text{agg}}(\cdot)$ is the one-layer fully connected network, $\text{avg}(x, y)$ denotes the x and y averaging operationand, and $|E_{:,j}^{(l)}|$ denotes the number of edges connecting node j at the lth iteration. To prevent the GNNs from oversmoothing problem [30], we average the initialized feature graph with the fully connected layer output at each iteration.

In Fig. 4, (25) is depicted by Steps 1 and 2, while (26) is depicted by Steps 3 and 4. As the color change in features in the FPA of Fig. 4, after one iteration, the neighboring targets at moments t and $t-1$ complete an information diffusion. As the number of iterations increases, the information diffuses more widely, creating a one-to-many relationship feature, shown as $n_i^{(l)}$, $n_j^{(l)}$, and $e_{i,j}^{(l)}$ of Step 4 in Fig. 4.

The structures of the OAE and OV are illustrated in Fig. 4. These modules include a fully connected layer network, a sigmoid, and a threshold filter. Specifically, the OAE generates an edge score matrix, indicating the likelihood of connected detections representing the same object. Due to constraints in the association (20), the Hungarian algorithm is employed for optimal matching using this matrix. An edge is activated if its score exceeds threshold (τ_E), and the corresponding node's detection box is considered even without a qualifying score. In the OV module, node features assess the reliability of the augmented object. If a node's confidence score surpasses τ_N, the enhancement is maintained; otherwise, it is discarded. In summary, OAE retrieves detection boxes with lower scores using the association results, whereas OV readjusts to dismiss the retrieved interferences. To enhance understanding of OAE, OV, and LSMcI, we delineate pseudocode in Algorithm 1.

D. Training Techniques

GMFTracker undergoes training utilizing both detection loss and association loss. This section delineates the detection loss

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function ($\mathcal{L}_D$), association loss function ($\mathcal{L}_A$), and methodology for assigning association labels, sequentially.

Our baseline model is FairMOT [14], while GMFTracker incorporates motion direction prediction compared with the baseline model. Thus, the detection loss is as follows:

$$\mathcal{L}_D = w_{\text{score}}\mathcal{L}_{\text{score}} + w_{\text{size}}\mathcal{L}_{\text{size}} + w_{\text{off}}\mathcal{L}_{\text{off}} + w_{\text{angle}}\mathcal{L}_{\text{angle}}. \tag{27}$$

$\mathcal{L}_{\text{score}}$, $\mathcal{L}_{\text{off}}$, and $\mathcal{L}_{\text{size}}$ are consistent with FairMOT specifications and will not be repeated here. Inspired by the rotating object detection task [46], we use the $L1$ loss to regress the motion angle.

Our association loss is computed by applying weights and summing the edge and node classification loss as follows:

$$\mathcal{L}_A = w_{\text{edge}}\mathcal{L}_{\text{edge}} + w_{\text{node}}\mathcal{L}_{\text{node}}. \tag{28}$$

In GMFTracker, the outputs of the OAE and OV are the node's score $\hat{s}_i^n \in S_V$ and the edge's score $\hat{s}_{i,j}^e \in S_E$, respectively. During the network training, we employ minimally constrained center distances to generate GTs for edge classification and node classification. When training of the node classifier, if the center of the detection box $\hat{b}_t^i$ is located within $\tau_{\text{ass}}(\tau_{\text{ass}} < 1)$ multiple of the corresponding GT box $b_{i'}$, the label of that node is designated as positive ($s_{i,i'}^n = 1$); when training of an edge classifier, if the labels for the ith and jth nodes within the same $b_{i'}$, and both are assigned positive values ($s_{i,i'}^n = 1$ and $s_{j,i'}^n = 1$), the labels of the edges linking these nodes are similarly assigned positive value ($s_{i,j}^e = 1$). The formula for calculating the loss is as follows:

$$\mathcal{L}_{\text{node}} = \frac{1}{|V_+^l|}\sum_{n_i^l \in V^l} \text{FL}\left(s_i^n, \hat{s}_{i,i'}^n\right) \tag{29}$$

$$\mathcal{L}_{\text{edge}} = \frac{1}{|E_+^l|}\sum_{e_{i,j}^l \in E^l} \begin{cases} \text{FL}\left(s_{i,j}^e, \hat{s}_{i,j}^e\right), & \text{if}\left(s_{i,i'}^n || s_{j,i'}^n = 1\right) \\ 0, & \text{otherwise} \end{cases} \tag{30}$$

where FL$(\cdot)$ denotes focal loss, addressing sample imbalance in the initial stages of training.

IV. Experiments

Within this section, we prove the efficacy of our suggested approach via a sequence of ablation studies and comparative analyses. Prior to this, we delineate the experimental setup, the evaluation metrics employed, and the dataset utilized.

A. Experimental Setup

1) Our GMFTracker: The proposed tracking system, based in FairMOT for joint detection and tracking, employs the SGD optimizer for model training, aligning with practices in [14]. The training spans 36 epochs, with an initial learning rate of $2e^{-4}$, undergoing a tenfold decay at the 24th and 33rd epochs. Each training iteration processes a batch of eight images, sourced from four randomly chosen video sequences. For the VISO dataset, to ensure fair comparisons with other networks, the original images, sized at 1024 × 1024, are fed directly into the network. The same approach is adopted for the CGSTL dataset, with network input images also maintained at a size of 1024 × 1024. For the OAE and OV, $\tau_E = \tau_N = \tau_{\text{ass}} = 0.5$. An NVIDIA RTX3090 with 24 G of memory is utilized for testing.

2) Comparison Methods: Deep learning-based MOT networks have undergone rapid development, with several classical online tracking networks such as FairMOT [14], DeepSORT [16], OC-SORT [21], ByteTrack [20], YOLOv7 [47] with Bot-SORT [48], CFTracker [3], Deform-DETR [23], MOTRv2 [22], YOLOx [49], GSDT [15], OKD-JDT [7], TGraM [10], SUSHI [31], StrongSORT [32], SMILEtrack [33], Dino [35], UCMCTrack [34], SFSORT [36], and BoostTrack [37].

B. Metrics

Our objective is to develop GMFTracker as an online multiobject tracker. To facilitate a comparison with established online multiobject trackers, we adopt the evaluation metrics outlined by MOTChallenge, selecting ten widely recognized metrics. These include multiobject tracking accuracy (MOTA), ID F1 score (IDF1), ID precision (IDP), ID recall (IDR), false positives (FPs), false negatives (FNs), identity switches (IDs), mostly tracked trajectories (MT), mostly lost trajectories (ML), and frames per second (FPS), as referenced from [14]. To ascertain the efficacy of the task alignment function and the enhanced association detection function, we utilize $\text{mAP}_{S/M/L}$, mAP_{50}, mAP_{75}, $\text{mAP}_{50:95}$, and F1scores for assessing detection performance, where the area divisions of S/M are 4 × 4 pixels and 8 × 8 pixels.

C. Dataset

This experiment is mainly tested on three benchmarks, which are VISO [1], CGSTL [2], and TinyPerson [52]. The detailed description of them is as follows.

1) VISO: The VISO dataset contains 17 730 optical remote sensing images, each sized 1024 × 1024, distributed across 47 video sequences. These videos are captured by the Jilin-1 satellite with a spatial resolution of 1 m. We use the first seven videos as the test set and the remaining 40 videos as the training set. The VISO dataset has a significant amount of bridge, tree, and building occlusion, as well as interference similar to the target response. Therefore, VISO is able to validate the ability of the model to maintain the target trajectory.

2) CGSTL: The CGSTL dataset contains three videos with a total of 1740 images, each sized 1024 × 1024. Each video fragment has 580 images. The videos, captured by the Chang Guang Optical Satellite with a spatial resolution of 1 m, only include car objects. To verify the generalization performance of the model, all CGSTL videos are used as the test set, while the 40 videos of VISO serve as the training set.

3) TinyPerson: The TinyPerson is a benchmark dataset of tiny objects captured by UAVs in real-world scenes. It contains 1610 images, with 794 used for training and 816 for testing. The objects in TinyPerson are very small, with 85% of the target areas less than 32 pixels, making the problem of localization drift even more serious. Therefore, TinyPerson can be used to validate the improvement of detection accuracy.

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TABLE I

ABLATION RESULTS FOR CE, SCCO, AND TFF IN DETECTION AND TRACKING TASKS. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER

DET							
CE	SCCO	TFF	mAP_S	mAP_M	mAP_L	mAP_{50}	F1 Score
			47.87	66.55	78.95	57.98	59.82
✓			49.94	69.59	83.77	60.07	61.93
	✓		48.15	68.87	81.96	59.02	60.78
✓	✓		52.24	72.84	84.36	60.51	64.13
✓	✓	✓	**53.31**	**74.16**	**86.21**	**61.78**	**65.57**
MOT							
CE	SCCO	TFF	MOTA↑	IDF1↑	IDs↓	MT↑	ML↓
			47.3	53.7	821	437	166
✓			47.6	54.1	796	444	**115**
	✓		47.9	53.8	757	419	154
✓	✓		48.7	54.8	754	459	125
✓	✓	✓	**49.3**	**56.1**	**731**	**469**	116

TABLE II

GAIN DUE TO TOTC IN DIFFERENT CLASSICAL DETECTORS

MODEL	TOTC	mAP_{50}	mAP_{75}	$mAP_{50:90}$	F1 Score	FPS
YOLOx		73.1	20.0	24.2	71.3	157
	✓	73.9	22.3	27.5(+2.7)	73.2(+1.9)	136
FCOS		51.8	15.6	20.5	53.6	177
	✓	53.9	18.8	22.9(+2.4)	54.8(+1.2)	158
Faster RCNN		67.3	16.9	21.5	60.3	148
	✓	68.9	20.1	24.9(+3.4)	62.4(+2.1)	127

TABLE III

RESULTS OF LSMCI ABLATION IN THE VISO DATASET. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER

OA	OE	OV	TOTC	MOTA↑	IDF1↑	IDs↓	MT↑	ML↓
				47.3	53.7	821	437	166
✓				47.7	55.4	719	441	130
✓	✓			48.2	56.1	667	456	103
✓	✓	✓		**50.3**	**59.5**	**627**	**476**	**95**
			✓	49.3	56.1	731	469	116
✓			✓	50.2	56.9	637	463	107
✓	✓		✓	50.8	57.3	612	455	96
✓	✓	✓	✓	**52.0**	**61.2**	**579**	**481**	**88**

D. Ablation Study

1) Effectiveness of TOTC: Table I presents the ablation results for the proposed TOTC. As delineated in [14], the baseline for the GMFTracker refers to the FairMOT without CE, SCCO, and TFF components, while its detection portion is very similar to CenterNet. To analyze the role of the TOTC, first we show the ablation results of the TOTC for object detection and MOT in the VISO dataset. Next, we performed ablation experiments on CE, SCCO, and TFF.

In the VISO dataset, the CE enhances the baseline performance of object detection with improved mAP_{50} and F1 scores. For tracking tasks, CE increases the MOTA and IDF1 scores relative to the object tracking baseline. Meanwhile, SCCO improves object detection with a 1.04 increase in mAP_{50} and a 0.96 rise in the F1 score. For object tracking, SCCO enhances MOTA by 0.6 and IDF1 by 0.1 compared with the baseline. TFF combines four-step temporal features, which facilitates motion target detection and motion direction regression. Therefore better detection accuracy and association accuracy are obtained when the TFF module is turned on. Fig. 5 exhibits the results of feature map visualization, highlighting the TOTC's impact on a scenario randomly selected from the VISO dataset. (a) and (c), as well as (b) and (d), delineate the state's pre-TOTC and post-TOTC application, respectively. (a) and (b), as well as (c) and (d), depict the feature map visualization outcomes for classification and localization tasks, correspondingly. A visualization technique called Grad-CAM [50] is used to visualize the feature maps of H_{hm}^{align} and $H_{\text{reg}}^{\text{align}}$. It is evident that prior to the implementation of TOTC, there is a significant misalignment among the classification feature map, the localization feature map and the true target, with strong response points dispersed amidst environmental disturbances. Subsequent to the application of the TOTC, a substantial improvement was obtained.

To ascertain the efficacy of TOTC in TOD, we incorporated TOTC into various classical detectors (see [28], [49], [51]) and verified generalizability in the TinyPerson dateset. To ascertain the generalizability of TOTC, no modifications are made to the original detector. As shown in Table II, the outcome of these experiments indicates that TOTC has a robust generalization capability.

2) Effectiveness of LSMcI: LSMcI comprises three operations: OA, OE, and OV. OA handles object association, OE emphasizes the restoration of low-confidence detection through association results, and OV performs an interference rejection operation on the target. To more precisely investigate each module's role, we sequentially ablate the aforementioned three modules in the VISO dataset.

Table III presents the ablation studies for LSMcI, and for a comparative analysis with TOTC, a series of tests are conducted both incorporating and excluding TOTC integration. Compared with the baseline model, integrating OA improved MOTA, IDF1, and MT while reducing IDs and ML. The modest rise in MOTA following OE's integration is due to OE's dual effect. Not only does it recover the actual target, but it also reinstates some of the interference. However, the incorporation of the interference rejection mechanism OV resulted in substantial improvements across all metrics: MOTA, IDF1, and MT increased, while IDs and ML decreased. It is evident that the integration of TOTC and LSMcI markedly enhances the tracking effect.

Furthermore, experiments are carried out to evaluate the impact of varying the number of detection boxes, K, entering the LSMcI. Table IV illustrates that the influence of different K values on inference outcomes follows a discernible trend. When K exceeds 200, there is negligible change in MOTA and IDF1, but the FPS begins to diminish significantly.

Subsequently, we conduct ablation studies on the number of GNN iterations to assess their impact on tracking performance. With the escalation in iteration count, each node gains the capability to amalgamate information from nonadjacent nodes that are more distant. Nonetheless, as indicated in [30], a higher iteration count does not unequivocally translate to superior outcomes. As evidenced by the data presented in

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TABLE IV

DIFFERENT TOPK MEASUREMENTS OF MOT PERFORMANCE AND SPEED. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER

TopK	MOTA↑	IDF1↑	IDs↓	MT↑	ML↓	FPS↑
100	48.3	56.7	746	428	110	**14.2**
200	**52.0**	**61.2**	**579**	**481**	**88**	13.7
300	51.8	61.0	621	479	89	13.0
400	51.8	61.0	625	479	91	12.4

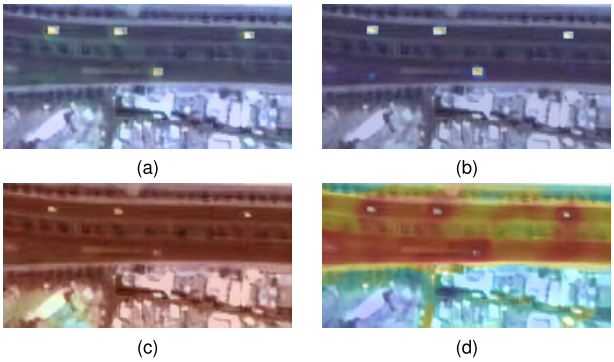


Fig. 5. Feature visualization results in the VISO dataset scenario before and after TOTC effect. (a) Classification features without TOTC. (b) Classification features with TOTC. (c) Localization features without TOTC. (d) Localization features with TOTC. (a) and (b) Visualizing GT with transparent boxes.

TABLE V

EFFECT OF THE NUMBER OF FPA ITERATIONS. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER

N_{iter}	MOTA↑	IDF1↑	IDs↓	MT↑	ML↓	FPS↑
2	50.5	60.1	624	488	96	**14.3**
3	52.0	61.2	579	481	88	13.7
4	52.1	61.5	563	483	87	13.2
5	**52.3**	**61.7**	**517**	**499**	**84**	12.7
6	50.8	60.7	593	496	89	12.0
7	49.9	59.9	662	485	103	11.2

Table V, optimal tracking performance is attained when the iteration count is 5. Beyond this point, more iterations cause oversmoothing in the GNNs.

Eventually, we also devise experiments to validate model generalizability, such as Fig. 6, to investigate the impact of clue fusion methods and the quantity of association clues on model generalizability. The same detector is employed for each experimental series, wherein the training set is from VISO and the test set is from CGSTL. We select average weight distribution, weight distribution guided by a priori knowledge, weight distribution guided by Bot-SORT, weight distribution guided by ByteTrack, and LSMcI for clues fusion. The experimental outcomes reveal that LSMcI's IDF1 consistently excels when the count of clues surpasses 2. It must be underscored that, apart from LSMcI, other cue fusion methods exhibit a declining trend in IDF1 scores upon integrating Sim cues. This phenomenon can be attributed to the reliability of Sim for tiny objects being confined to neighboring targets.

E. Comparisons With Other Trackers

1) Quantitative Analysis: In the VISO dataset, the GMFTracker achieves leading tracking performance with

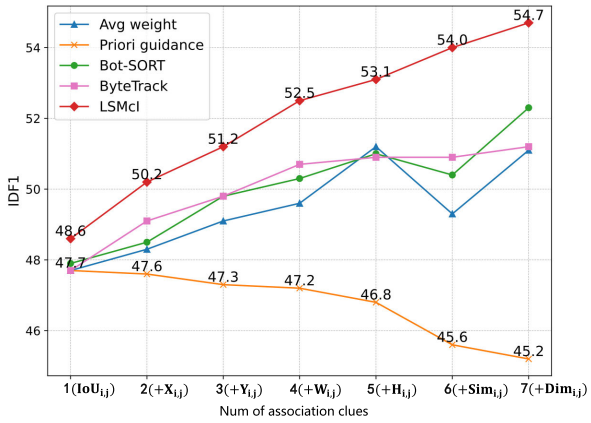


Fig. 6. Experiments on model generalizability validation. Varied colored lines represent distinct fusion methods, while the horizontal axis denotes the quantity and identity of combined clues, as delineated in (21).

adequate inference speed. Table VI displays its superior performance, showing the best and second-best outcomes. In particular, GMFTracker achieves optimal performance metrics in MOTA, IDF1, MT, and ML, while demonstrating secondary-level outcomes in IDR, IDP, FN, and IDs. The achievement of optimal MOTA is attributed to GMFTracker's TOTC. By means of task alignment, this ensures that detection boxes with the highest confidence scores simultaneously produce the most advantageous localization outcomes. The FN and FP metrics perform remarkably well, attributable to the LSMcI's association enhancement and interference rejection capabilities. The optimal achievement in IDF1 is attributed to LSMcI's capability to integrate association clues from neighboring targets and dynamically determine the importance of these clues. Both MT and ML metrics attain optimal results, indicating GMFTracker's proficiency in sustaining target tracking overextended durations. This observation is corroborated by the findings presented in Fig. 7, which illustrate GMFTracker's occlusion resistance. Notably, the FPS of our proposed tracker attains 12.7, surpassing the transmission frame rate of satellite video, thereby facilitating online tracking tasks.

Table VII presents the performance outcomes of GMFTracker on the CGSTL dataset, with the highest scores in each metric highlighted in bold. To assess the generalization capabilities of all trackers, training on CGSTL data was deliberately omitted. Consequently, the VISO dataset is employed as the training set, whereas the testing is exclusively conducted on CGSTL. Table VII illustrates that the GMFTracker we developed achieves optimal performance in MOTA, IDF1, IDR, MT, ML, and FN, while it records secondary-level results for IDs. In intricate satellite video contexts, it is not feasible to ensure that a weighted amalgamation of various clues will address the association challenge across all scenarios. The node-relationship characteristic of GMFTracker, which is adaptable to the surrounding environment, eliminates the need for incorporating a priori knowledge. This significantly contributes to GMFTracker's distinctly superior generalization compared with other trackers.

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Fig. 7. Examples of detections and trajectories. (a) and (b) Tracing outcomes in challenging scenes from the VISO and CGSTL, respectively. Specifically, the green box represents the GT, the blue box signifies the output result, the red box indicates a false alarm, and identical target IDs share the same track color. Evidently, our approaches yield more precise and complete trajectory plots in challenging scenarios.

2) Qualitative Analysis: Fig. 7(a) and (b) presents a comparative visualization between GMFTracker and other prevailing trackers in satellite videos. Selected challenging scenes from VISO and CGSTL datasets are intentionally chosen for this comparison. The visualization of the detection results in (a) and (b) highlights our method furnishes superior interference suppression ability, enabling it to differentiate between street lights and lane lines, which exhibit similar responses to target pixels. This capability is attributed to the OV module. Furthermore, the first line in (b) reveals that our OE module can reconstruct detection boxes through edge activation results, despite its low confidence level. The visualization results of the trajectories illustrate our method's proficiency in reducing ID switches, owing to the OE module's dynamic evaluation of various clues' significance. The second line in (b) shows interbuilding tracking results and demonstrates the long-term tracking capability and interference suppression of our method.

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TABLE VI

QUANTITATIVE RESULTS IN VISO'S TEST SET. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER. THE BEST RESULTS ARE SHOWN IN BOLD RED, AND THE SECOND-BEST RESULTS ARE SHOWN IN BOLD BLACK. † INDICATES A GRAPH-BASED MODELS

Method	Detector	MOTA↑	IDF1↑	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	IDs↓	FPS↑
FairMOT [14]	CenterNet	21.4	13.8	**76.8**	7.6	33	1021	**1950**	92304	**26**	**21.2**
Deep SORT [16]	YOLOx-X	44.5	47.3	46.5	48.1	376	355	28457	34244	3871	**20.7**
ByteTrack [20]	YOLOx-X	44.3	58.2	64.7	53.0	343	126	28299	34794	3719	19.6
OC-SORT [21]	YOLOv7-X	48.8	58.7	61.5	56.2	466	129	25620	35316	578	17.9
Bot-YOLOv7 [48]	YOLOv7-X	46.1	48.3	60.3	56.5	275	235	26457	35225	2971	5.4
GSDT† [15]	DSFNet	48.1	47.9	45.9	50.1	291	313	24145	34981	3128	6.7
SUSHI† [31]	Cascade-RCNN	50.2	55.6	60.4	51.5	**489**	98	24032	35108	593	8.1
TGraM† [10]	Swin-s+STeRe	49.1	58.3	60.1	56.6	332	147	27663	**32695**	697	16.8
CFTracker [3]	CFU	**50.9**	-	-	-	392	100	23542	34684	641	7.8
OKD-JDT [7]	-	49.9	48.2	65.7	38.0	329	293	**1949**	54120	5828	-
MOTRv2 [22]	Deform-DETR	49.6	**60.2**	63.5	**57.4**	345	114	24651	35196	607	11.3
SMILEtrack [33]	PRB-Net	39.4	45.9	42.5	49.9	243	466	29132	38890	4667	20.4
UCMCTrack [34]	YOLOv7-X	47.1	51.0	53.7	48.7	288	396	24947	34988	3519	11.6
SFSORT [36]	Swin-b+Dino	49.1	56.3	59.5	53.4	347	178	24750	35203	1101	8.3
StrongSORT [32]	Swin-b+Dino	48.9	57.2	59.8	54.9	398	**93**	24955	35578	761	8.9
BoostTrack [37]	Swin-b+Dino	48.7	53.6	55.5	51.9	377	334	24158	35680	1696	9.2
GMFTracker(Ours)	TOTC	**52.3**	**61.7**	**66.9**	**57.3**	**499**	**84**	23466	**33231**	**517**	12.7

TABLE VII

QUANTITATIVE RESULTS IN CGSTL. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER. THE BEST RESULTS ARE SHOWN IN BOLD RED, AND THE SECOND-BEST RESULTS ARE SHOWN IN BOLD BLACK. † INDICATES A GRAPH-BASED MODELS

Method	Detector	MOTA↑	IDF1↑	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	IDs↓
Deep SORT [16]	YOLOx-X	32.9	46.1	58.3	38.1	44	50	**6225**	19453	314
ByteTrack [20]	YOLOx-X	32.7	47.1	58.4	39.4	45	49	6737	19102	231
OC-SORT [21]	YOLOv7-X	33.6	52.1	**64.0**	43.9	52	48	6714	18799	208
Bot-YOLOv7 [48]	YOLOv7-X	31.2	51.1	61.2	43.8	50	43	7753	18687	211
GSDT† [15]	DSFNet	32.3	50.6	60.3	43.6	45	49	6859	19366	245
TGraM† [10]	Swin-s+STeRe	34.4	51.4	61.6	44.1	48	45	6683	18499	229
SUSHI† [31]	Cascade-RCNN	35.7	**53.1**	63.7	**45.5**	**54**	**40**	6779	17954	**174**
MOTRv2 [22]	Deform-DETR	33.9	49.3	59.4	44.2	47	48	6869	18538	198
SMILEtrack [33]	PRB-Net	29.7	47.1	54.3	41.7	31	57	7101	19654	478
UCMCTrack [34]	YOLOv7-X	34.9	49.5	58.9	42.7	46	47	6707	18277	234
SFSORT [36]	Swin-b+Dino	**35.8**	52.4	**63.9**	44.4	51	42	6831	**17844**	194
StrongSORT [32]	Swin-b+Dino	35.6	52.5	63.0	45.1	50	41	6843	17901	203
BoostTrack [37]	Swin-b+Dino	35.7	51.3	62.3	43.7	43	46	6802	17865	241
GMFTracker(Ours)	TOTC	**40.3**	**54.7**	63.1	**48.3**	**61**	**33**	**6613**	**16332**	**179**

TABLE VIII

COMPARISON OF MODEL EFFICIENCY. MULTI-ADDS AND FPS ARE ESTIMATED ON AN NVIDIA RTX 3090 GPU

Method	Backbone	Params	Multi-adds	Time(ms)
FairMOT [14]	DLA34	20.0M	72.2G	47.2
SUSSHI [31]	R-101	53.5M	158.6G	123.5
CFTracker [3]	DLA34	-	-	128.2
TGraM [10]	Swin-s	59.4M	113.1G	59.5
MOTRv2 [22]	Swin-s	66.7M	122.8G	88.5
StrongSORT [32]	Swin-b	97.6M	207.1G	112.4
GMFTracker (DOL off)	DLA34	22.8M	98.3G	89.3
GMFTracker (DOL on)	DLA34	23.5M	83.3G	78.7

F. Efficiency

Table VIII compares the inference efficiency of representative models, tested with 100 randomly selected images from the VISO dataset. From Table VIII, GMFTracker is the second most efficient after TGraM. Its performance in inference efficiency is mainly due to the DOL in TOTC; without DOL, H_{reg} would need to compute the similarity with the global features to correct the localization features, which would significantly increase the computational overhead.

G. Conclusion and Future Work

This article introduces an online end-to-end multiobject tracker called GMFTracker. To address the problem of localization drift in TOD, GMFTracker is equipped with a TOTC module. TOTC can compensate the feature offsets between classification and localization tasks in the learning phase to obtain accurate localization results. Meanwhile, DOL uses the idea of sparse sampling to promote the inference speed of the GMFTracker. In the OA stage, prevalent models depend on a priori knowledge, often concentrating solely on one-to-one associations, overlooking one-to-many relationships. LSMcI utilizes FPA to pass and aggregate relational features between neighboring targets to obtain accurate association results without any a priori assumptions. Meanwhile, OE and OV accomplish OE and interference rejection operations, respectively, to more accurately maintain the target's trajectory. Extensive experiments on the VISO, CGSTL, and TinyPerson datasets validate the superior performance of the proposed tracker.

Nevertheless, GMFTracker underperforms in prolonged occlusion scenarios (>40 frame), particularly when tracking objects obscured by a dual-direction traffic overpass. Future research will aim at improving robustness in prolonged occlusion scenarios by deducing the logic of target disappearance and reappearance from object behavior and temporal data.

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